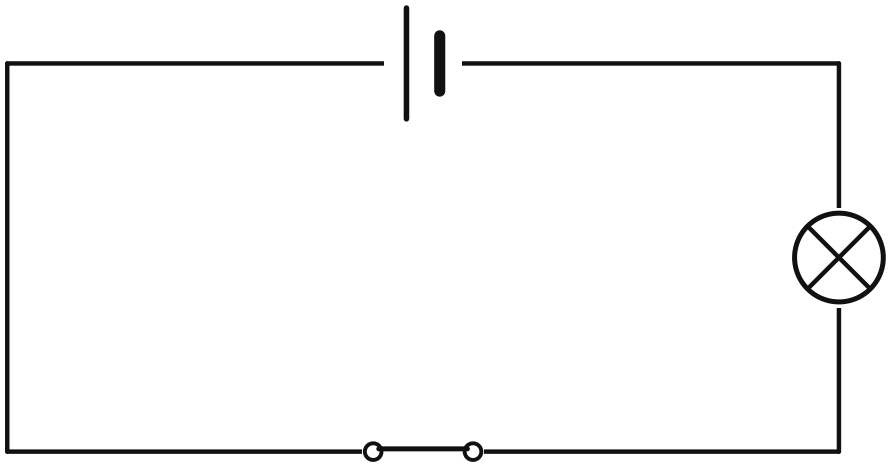
PHYSICAL AND CHEMICAL SCIENCE

Electricity and Circuits

Complete circuits, symbols, conductors and insulators, changing circuits, switches and safety



In this pack

- **Learn it:** Electricity and circuits ★☆☆
- **Fact check:** Circuits and symbols ★☆☆
- **Think it through:** Predict and explain ★★☆
- **Investigate:** Does wire length change brightness? ★★☆
- **Real-world case study:** Lights for every home ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Electricity and circuits

Name: Class: Date:

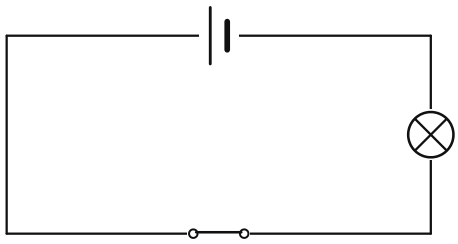
Learning goals

- Build and draw series circuits using conventional symbols.
- Sort materials into conductors and insulators and explain how they are used in cables and plugs.
- Predict how changing a circuit changes it, and explain why switches and safety devices matter.

Press a switch and the fan starts spinning. Plug in a phone and it charges. Electricity powers our homes, schools, hospitals and factories. But how does it get from the battery or the wall to the bulb? It needs a complete path called a **circuit**.

A complete circuit

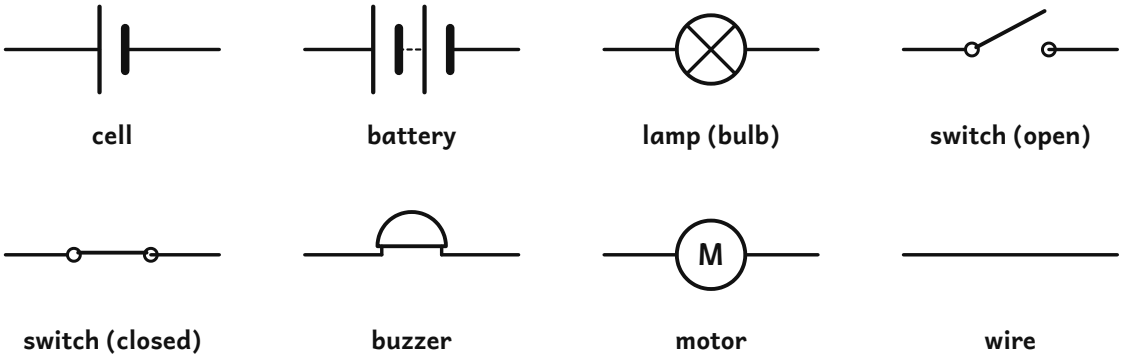
Electricity flows from one end of a **cell** (battery), through **wires** and **components** such as a bulb, and back to the other end of the cell. The path must be **complete**, with no gaps. If there is a gap, the electricity stops and the bulb goes out. A circuit where everything is joined in one loop is a **series circuit**.



A series circuit with a cell, a lamp and a closed switch.

Circuit symbols

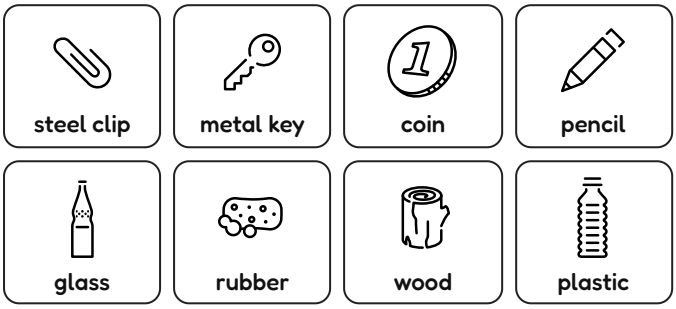
Scientists all over the world draw circuits with the same simple **symbols**, so everyone can understand them.



Circuit symbols.

Conductors and insulators

Conductors let electricity flow through them easily. Most **metals** are good conductors: copper, aluminium, iron, steel, silver. Graphite (pencil "lead") conducts too. **Insulators** do not let electricity flow: plastic, rubber, wood, glass, dry paper and cloth.



Electrical **cables** have **copper** wires inside, because copper is an excellent conductor. They are covered in **plastic** to stop electricity reaching our hands. Plugs and switches have plastic cases for the same reason.

Changing a circuit

Change	What happens to the bulb
add another cell	brighter
add another bulb	each bulb is dimmer
use a longer wire	a little dimmer
use a thinner wire	a little dimmer
open the switch	goes out
swap the bulb for a buzzer or motor	it buzzes or spins

Switches and safety

A **switch** makes a gap in the circuit (off) or closes it (on), so we can control devices. **Fuses** and **circuit breakers** are safety switches: if too much electricity flows, they cut it off before wires get hot and start a fire.

> ⚠ **Stay safe**
>
> **Mains electricity** from the wall socket (220 volts in Bangladesh) is about 150 times stronger than a 1.5 volt cell and can kill. Never touch sockets or switches with wet hands, never push anything into a socket, do not plug too many things into one socket, and never fly kites near power lines.

Electricity all around us

At **home**: lights, fans, fridges, pumps, phone chargers. At **school**: computers, projectors, bells. In the **community**: traffic lights, hospital machines, water pumps, street lights and mobile phone towers. Each is an **electrical system**: a source, wires, switches and components working together.

Key words

circuit	a complete path for electricity
cell (battery)	a store of energy that pushes electricity round a circuit
component	a part of a circuit, such as a bulb or switch
conductor	lets electricity flow
insulator	does not let electricity flow
switch	opens or closes a gap in a circuit
series circuit	all components in one loop
mains electricity	electricity from the wall socket

FACT CHECK · C6 · ELECTRICITY

Circuits and symbols

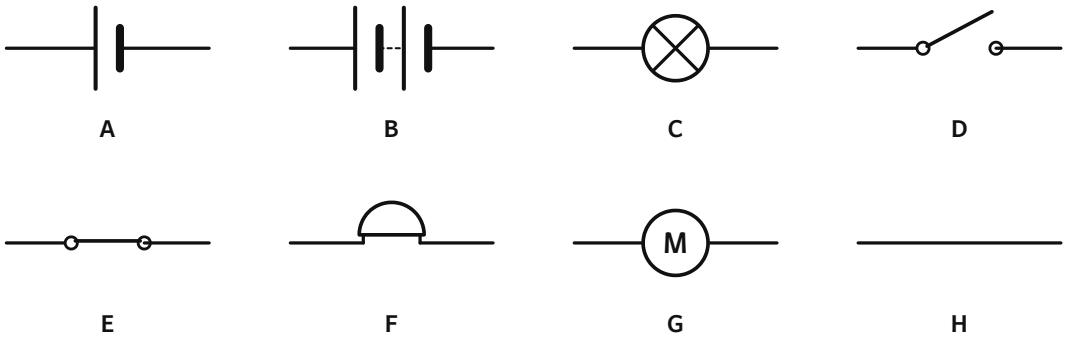
★☆☆
Grade 4

Name: Class: Date:

Learning goals

- Use circuit symbols.
- Sort conductors and insulators.

1 Name each symbol. [8]



A B C D

E F G H

A	B	C	D	E	F	G	H

2 Sort the materials. [8]

SORT THESE

wooden stick **pencil lead (graphite)** **steel spoon** **copper** **glass**
aluminium foil **rubber band** **plastic ruler**

Conductor	Insulator

3 **Will the bulb light? Write yes or no and explain.** [4]

Circuit	Lights?	Why?
cell, bulb and wires in a complete loop		
one wire is not connected		
a plastic straw joins the gap		
a metal key joins the gap		

THINK IT THROUGH · C6 · ELECTRICITY

Predict and explain

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Predict changes to circuits.
- Explain safety features.

1 **Draw a circuit diagram with two cells, one lamp and a switch. Use the correct symbols.** [3]

My circuit

2 **Predict and explain.** [6]

Change	Prediction	Reason
add a second bulb		
add a second cell		
make the wire much longer		

3 **Why is the inside of a cable made of copper and the outside made of plastic?** [2]

4 **Why is it dangerous to use a hair dryer with wet hands in the bathroom?** [1]

Does wire length change brightness?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Plan a fair test.
- Measure with a light meter.
- Draw and use a line graph.

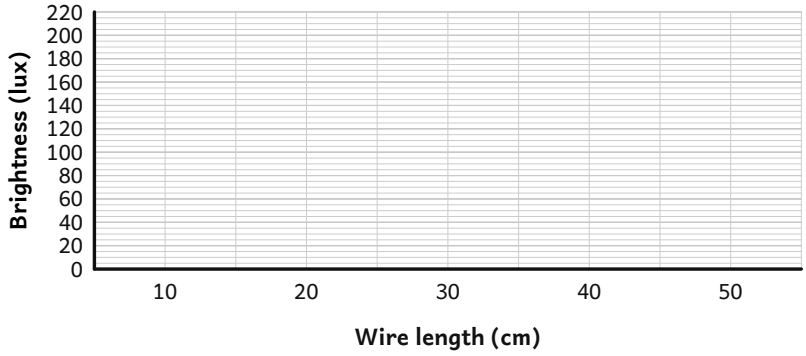
Class 4C made a circuit with a cell, a bulb and a long piece of thin nichrome wire. They moved a crocodile clip along the wire to change the length in the circuit, and used a light meter app to measure the brightness of the bulb in lux.

1 **Complete the plan.** [3]

Change	
Measure	
Keep the same	

2 **Draw a line graph.** [3]

Wire length (cm)	10	20	30	40	50
Brightness (lux)	200	160	130	110	95



3 **Describe the pattern.** [2]

.....

.....

4 **Predict the brightness for 60 cm.** [1]

.....

5 **A dimmer switch for a lamp works in a similar way. How do you think it works?** [1]

REAL-WORLD CASE STUDY · C6 · ELECTRICITY

Lights for every home

★★★ Grade 4

Name: **Class:** **Date:**

Learning goals

- Use data about electricity in Bangladesh.
- Think about electrical safety.
- Plan how to save electricity.

Science · History · Maths · Citizenship

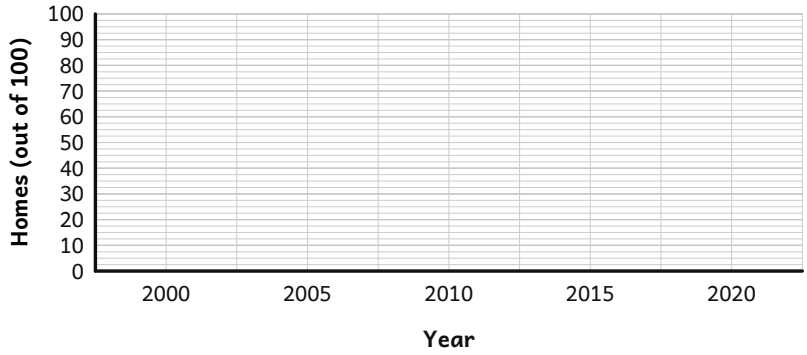
In the year 2000, only about **one in three** homes in Bangladesh had electricity. Children studied by the light of kerosene lamps, which were smoky and dangerous. Since then, new power stations, power lines and millions of **solar home systems** have brought electricity to almost **every home** in the country.

Electricity has changed lives: children can study at night, farmers use electric pumps, shops stay open and people charge mobile phones. But more electricity also means more need for **safety** and for **saving energy**, because much of it is made by burning gas and oil.

Approximate figures from World Bank data on access to electricity in Bangladesh.

1 **Homes with electricity in Bangladesh (approximate).** [3]

Year	2000	2005	2010	2015	2020
Homes with electricity (out of 100)	32	44	55	74	96



a) Draw the line graph.

b) By how much did it rise from 2000 to 2020?

.....

2 **Give two ways electricity has helped families.** [2]

.....

.....

3 **Spot the danger. What is wrong in each case and how should it be fixed?** [3]

Situation	Danger and fix
six plugs in one socket with an adaptor	
a cable with the plastic cover cracked	
a kite caught on a power line	

4 **Write three tips for saving electricity at home.** [2]

Show what you know

★★★ Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

A	C	W	N	K	B	J	I	W	P
V	J	O	S	I	U	R	F	I	L
L	E	T	P	G	Z	R	S	R	A
C	F	C	F	P	Z	C	W	E	S
I	I	B	E	E	E	R	I	C	T
M	E	R	U	L	R	R	T	C	I
O	G	F	C	L	L	K	C	V	C
T	D	P	D	U	B	O	H	O	Y
O	E	L	O	M	I	H	A	U	S
R	U	K	O	D	A	T	C	I	I

bulb buzzer cell circuit copper motor plastic switch wire

2 Which material is a conductor? [1]

- ◯ rubber
- ◯ plastic
- ◯ copper
- ◯ wood

3 What happens if you add a second bulb to a series circuit? [1]

- ◯ both bulbs are brighter
- ◯ both bulbs are dimmer
- ◯ nothing changes
- ◯ the cell explodes

4 Design a burglar alarm for a box: when the lid opens, a buzzer sounds. Draw the circuit. [3]

Box alarm

5 **Challenge. Why is it important to have *a* switch in *a* circuit for *a* motor that turns *a* fan?** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
build and draw *a* series circuit	○	○	○
use circuit symbols	○	○	○
sort conductors and insulators	○	○	○
predict changes to *a* circuit	○	○	○
investigate wire length and brightness	○	○	○
stay safe with electricity	○	○	○